

research in power controls is that each node cooperates with all other nodes in selecting the individual transmission power level.

Smart sensors

The IEEE 1451 standard makes it easier for different manufacturers to develop smart sensors and to interface these devices to networks. A smart sensor is a sensor that provides extra functions beyond those necessary for generating a correct representation of the sensed quantity. These include signal conditioning, signal processing and decision making/alarm functions.

Main objectives of smart sensors are: (a) move the intelligence closer to the place of measurement, making it cost-effective to integrate and maintain distributed sensor systems, (b) clustering of transducers, (c) control, compute and communicate for a common cause and (d) seamlessly interface many sensors of different types. The concept of virtual sensor is also fulfilled.

A virtual sensor, a component of smart sensor, is a sensor/transducer plus the associated signal conditioning and DSP. MEMS sensors are now well developed and are available for most sensing applications in WSNs.

Transduction mechanisms of sensors

Transducers convert energy from one domain to another and their outputs are generally voltages or currents. The sensory transduction may be carried out by using any one or more of the following effects of physical principles: piezo-resistive, piezo-electric, capacitive, inductive, magnetic or electro-magnetic, magneto-resistive, magnetic field, thermal (thermo-mechanical, thermo-resistive, thermocouples, resonant temperature sensors, thermopiles or bolo meters).

Optical transducers. These make use of photo-electric effect, photo-conductive effect, or optical-fibre based technology (for instance, devices like accelerometers often use time-of flight information, IR detectors or UV detectors).

Electro-chemical, chemical and biological. Used for solids, liquids and gases, these include metal-oxide gas sensors, interdigitated-gate electrode FETs (IGFETs) and their arrays, ion-sensitive FETs (ISFETs) like ionophone, quartz crystal micro-balance (QCM) and micro-cantilevers. Types of gases that can be monitored using metal-oxide gas sensors are CO_2 , CO , H_2S and NH_3 . Other applications are in medical diagnosis of smallpox, anthrax chemical warfare, food monitoring, security and surveillance (using implantable bio-sensors).

Acoustic wave sensors. Such sensors as surface acoustic wave (SAW), thickness shear mode (TSM) and flexural plate wave (FPW) make property changes, such as mass into detectable electrical signals.

Commercially available nodes come with five sensors installed for temperature, light, acoustic, acceleration/seismic and magnetic effects. Especially suitable for surveillance purposes, these work with operating frequency of 916MHz or 433MHz, with a data rate of 40k bits/sec, having a range of 10 to 30 metres. Installation of these devices requires a great deal of programming.

Ultra wide band (UWB) is of great interest in distributed networks, since UWB is a short-range technology that can penetrate walls. It is suitable for multi-node transmissions and has built-in time-of-flight properties that make it very easy to measure ranges down to 1cm with a range of 40m.

Moreover, UWB transceivers can be made very small and are amenable for MEMS technology. Since in pulse position modulation no carrier is needed, it means the antennae are not inductive. But signals coming from MEMS may be very noisy, of low amplitude, biased and dependent on secondary parameters, such as temperature, which is an undesired characteristic. That is why the signal conditioning circuits are very essential. ●

Concluded

The author is director (R&D) at Global Institute of technology, Jaipur

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